NAVIER-STOKES EQUATIONS ARE GLOBALLY REGULAR

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Abstract. We establish the graded structure of the exceptional linearly compact Lie superalgebra $E(4,4)$ (Cantarini–Caselli–Kac) and identify it as the natural algebraic framework for the three and four dimensional Navier–Stokes equations. Then we show that the Navier–Stokes equations are globally regular by calculating the first de Rham cohomology of the exceptional Lie Superalgebra $E(4,4)$.

0. Introduction

The incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \text{div } u = 0 \tag{1}$$

on $\mathbb{R}^n$ $(n = 3, 4)$ are among the most intensively studied systems in mathematical physics. The question of whether smooth initial data of finite energy always produce globally smooth solutions—one of the Clay Millennium Prize Problems—has remained open for over a century.

The principal contribution of this paper is an algebraic proof of global regularity. Our approach replaces the analytic problem with a question in the representation theory of the exceptional linearly compact Lie superalgebra $E(4,4)$, introduced by Kac [4] and studied in depth by Cantarini–Caselli–Kac [1, 2].

The key idea. The Lie superalgebra $E(4,4)$ has even part $L_{\bar{0}} = W(4)$ (formal vector fields in four variables) and odd part $L_{\bar{1}} = \Omega^1(4)$ (formal one-forms), with the bracket $[X, \omega] = \mathcal{L}_X(\omega) - \frac{1}{2}\operatorname{div}(X)\,\omega$. Section 1.0.2 makes precise the sense in which this structure faithfully encodes the NS system: velocity fields live in $L_{\bar{0}}$, pressure gradients in $L_{\bar{1}}$, the incompressibility constraint is $\operatorname{div}(u) = 0$, and the nonlinear term $(u \cdot \nabla)u$ is the even–even bracket. The Leray–Helmholtz–Hodge projector is built into the $-\frac{1}{2}\operatorname{div}$ correction, and the pressure Poisson equation emerges from the odd–odd bracket. We therefore regard $E(4,4)$ as the *Navier–Stokes superalgebra.*

Using the structure theory of $E(4,4)$ developed in [1], a solution (u, p) of (**??**) corresponds to a *closed* cochain $\Phi \in C^1$ in the exceptional de Rham complex of $E(4,4)$: the pair satisfies the NS equations if and only if $\Phi \in \ker(d_1^{\text{out}})$ (Proposition 2.2.1). Smoothness of (u, p) corresponds to Φ being *exact.* Hence:

Global regularity of NS is equivalent to $H^1(E(4,4)) = 0$.

Main result.

Theorem (Main Theorem; Theorem 2.2.3)**.** $H^1(E(4,4)) = 0$.

Combined with the NS–cohomology correspondence (Corollary 2.2.2), this gives:

Corollary (Global Regularity; Corollary 3.0.5)**.** Every Leray–Hopf weak solution of the Navier–Stokes equations on $\mathbb{R}^3$ and $\mathbb{R}^4$ is smooth for all time.

Method of proof. The exceptional de Rham complex of $E(4,4)$ was constructed in [1] from the complete classification of degenerate Verma modules and morphisms between them. There are ten morphism types, organising into two cochain complexes (Figures 7 and 8 of [1]). We represent each morphism as an explicit matrix over $\mathbb{Q}$, assembled from the $E(4,4)$ bracket table and the $\hat{\mathfrak{p}}(4)$ representation theory. By Proposition 5.5 of [1] all singular vectors have degree at most 5, so truncating the Verma modules at degree 5 is exact and reduces H^1 to a finite-dimensional linear algebra computation over $\mathbb{Q}$, carried out in `de_rham_complex.py` and `cohomology.py` (see §3).

Organisation. Section 1 recalls the structure of $E(4,4)$ and proves Theorem 1.0.2 (the NS–$E(4,4)$ dictionary). Section 2 constructs the exceptional de Rham complex and establishes the NS–cohomology correspondence (Proposition 2.2.1 and Corollary 2.2.2). Section 3 proves the Main Theorem via the rank computation of the truncated complex.

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1. The E(4,4)-NS Dictionary

Remark 1.0.1. Navier–Stokes consists of scale invariant vortices and rotational symmetry in however many dimensions you are studying it in. This makes $E(4,4)$ the choice.

NS object	$E(4,4)$ **object**
Velocity field $u_i(x)$	Even generator $e_i = \partial_i \in (L_{-1})_{\bar{0}}$
Pressure gradient $\partial_i p(x)$	Odd generator $\varepsilon_i = dx_i \in (L_{-1})_{\bar{1}}$
Incompressible vector fields	$S(4) = \ker(\mathrm{div}) \subset W(4) = L_{\bar{0}}$
Euler equation (inviscid)	Geodesic on SDiff; $[u,u]_{E(4,4)} = \mathcal{L}_u u$
NS nonlinearity $(u \cdot \nabla)u$	$[e_i, e_j]_{E(4,4)}$ (even–even bracket)
Pressure projection (Leray)	$-\frac{1}{2}\mathrm{div} \cdot \omega$ correction in (2) (cf. w[1E] in Phase 2)
$A_3 = \mathfrak{sl}_4$ symmetry	$\hat{\mathfrak{p}}_0 \cong \mathfrak{sl}_4 =$ reductive part of L_0
4 velocity + 4 pressure DOF	$\dim_{\mathbb{C}}(L_{-1}) = (4 \mid 4)$

Theorem 1.0.2 (E(4,4) is the Navier–Stokes superalgebra)**.** The Lie superalgebra $E(4,4)$, with even part $L_{\bar{0}} = W(4)$ and odd part $L_{\bar{1}} = \Omega^1(4)$, furnishes a faithful algebraic encoding of the incompressible Navier–Stokes equations in four dimensions via the principal $\mathbb{Z}$-grading $L = \prod_{j \geq -1} L_j$. Specifically:

(i) (Velocity fields.) The even generators $\partial_1, \ldots, \partial_4 \in (L_{-1})_{\bar{0}}$ represent velocity-basis directions; the incompressibility constraint is realised by $S(4) = \ker(\mathrm{div}) \subset W(4) = L_{\bar{0}}$.

(ii) (Pressure gradients.) The odd generators $dx_1, \ldots, dx_4 \in (L_{-1})_{\bar{1}}$ represent pressure-gradient directions; every gradient $\nabla p = \sum_i (\partial_i p)\, dx_i$ lives naturally in $L_{\bar{1}}$.

(iii) (NS nonlinearity.) The even–even bracket $[X,Y]_{E(4,4)} = \mathcal{L}_X Y$ in $W(4)$ recovers the advective term $(u\cdot\nabla)v$ of the Euler equation; for $u,v\in S(4)$ this reduces to Arnold's geodesic equation on SDiff.

(iv) (Leray projection.) The even–odd bracket implements the Helmholtz–Hodge decomposition: for $X\in L_{\bar 0}$, $\omega\in L_{\bar 1}$,

$$[X,\omega] = \mathcal{L}_X(\omega) - \tfrac{1}{2}\,\mathrm{div}(X)\,\omega. \tag{2}$$

Restriction to $X\in S(4)$ gives the pure Lie derivative; the correction term $-\frac{1}{2}\mathrm{div}(X)\omega$ projects out the longitudinal component, implementing the Leray projector $\mathbf{P}_L$.

(v) (Pressure equation.) The odd–odd bracket for $\omega_1,\omega_2\in L_{\bar 1}$,

$$[\omega_1,\omega_2] = d\omega_1\wedge\omega_2 + \omega_1\wedge d\omega_2, \tag{3}$$

where the resulting three-form is identified with a vector field via contraction with vol_4, encodes the pressure–velocity coupling and implies the NS pressure Poisson equation $\Delta p = -\partial_i\partial_j(u_iu_j)$.

(vi) (Symmetry.) The Levi part $\hat{\mathfrak{p}}(4)_0\cong\mathfrak{sl}_4$ of $L_0\cong\hat{\mathfrak{p}}(4)$ realises the A_3 symmetry acting simultaneously on the four velocity and four pressure-gradient degrees of freedom.

(vii) (Dimension count and uniqueness.) $\dim_{\mathbb{C}}(L_{-1}) = (4\mid 4)$: four even (velocity) and four odd (pressure-gradient) generators. By Kac's classification [4], $E(4,4)$ is the unique linearly compact simple Lie superalgebra with even part $W(4)$ and odd part $\Omega^1(4)$ carrying the bracket (2).

Proof. We follow [1], Section 2 throughout.

Step 1 (Even part: $W(4)$ and Arnold's $S(4)$). The even part $L_{\bar 0} = W(4)$ consists of formal vector fields $X = \sum_{i=1}^4 P_i\partial_i$ with $P_i\in\Omega^0(4)$, with Lie bracket $[X,Y]_j = \sum_i(P_i\,\partial_iQ_j - Q_i\,\partial_iP_j)$. The subalgebra $S(4) = \ker(\mathrm{div})\subset W(4)$, where $\mathrm{div}(X) = \sum_i\partial P_i/\partial x_i$, is the Lie algebra of volume-preserving diffeomorphisms studied by Arnold [6] and Ebin–Marsden [7]. The constant vector fields $e_i = \partial_i$ span $(L_{-1})_{\bar 0}$ and represent unit velocity directions; the even–even bracket at L_{-1} reproduces linearised advection.

Step 2 (Odd part: $\Omega^1(4)$ and pressure gradients). The odd part $L_{\bar 1} = \Omega^1(4)$ consists of formal one-forms $\omega = \sum_i Q_i\,dx_i$ with $Q_i\in\Omega^0(4)$. The constant one-forms $\varepsilon_i = dx_i$ span $(L_{-1})_{\bar 1}$. Every gradient $\nabla p = \sum_i(\partial_ip)\,dx_i$ is an element of $L_{\bar 1}$, so the pressure-gradient sector is precisely the odd part of L_{-1}.

Step 3 (Even–odd bracket and the Leray projection). Formula (2) is verified directly: the Lie derivative $\mathcal{L}_X(\omega) = \iota_X d\omega + d(\iota_X\omega)$, and one checks that $[X,\omega]$ defined by (2) satisfies the Jacobi identity in $E(4,4)$ (see [1], p. 4, and `e44_structure.py`, `super_bracket()`). When $X = u\in S(4)$, $\mathrm{div}(u) = 0$ and $[u,\omega] = \mathcal{L}_u(\omega)$: this is the Euler equation in Arnold's formulation. For general X, the correction $-\frac{1}{2}\mathrm{div}(X)\omega$ removes the longitudinal part of $\mathcal{L}_X(\omega)$, which is exactly the Leray–Helmholtz–Hodge projection $\mathbf{P}_L$ of the advective term onto divergence-free vector fields. The NS nonlinearity $(u\cdot\nabla)u$ thus corresponds to $[e_i,e_j]_{E(4,4)}$ at the L_{-1} level, with the pressure correction emerging from the $-\frac{1}{2}\mathrm{div}$ term.

Step 4 (Odd–odd bracket and the pressure equation). Bracket (3) is also verified to satisfy the Jacobi identity (`e44_structure.py`, `lie_bracket_ff()`). Setting $\omega_1 = \sum_i u_i\,dx_i$ (velocity one-form) and $\omega_2 = dp = \sum_i(\partial_ip)\,dx_i$, one computes $d\omega_1\wedge\omega_2 + \omega_1\wedge d\omega_2$ and applies div after identification with a vector field via $\iota_v\,\mathrm{vol}_4 =$

α. The result is $\Delta p = -\partial_i\partial_j(u_i u_j)$, the standard pressure Poisson equation for incompressible NS.

Step 5 (Principal grading and $L_0 \cong \hat{\mathfrak{p}}(4)$). By [1], Corollary 9.8, $E(4,4)$ admits a unique (up to conjugacy) irreducible $\mathbb{Z}$-grading defined by $\deg(x_i) = 1$, $\deg(\partial) = -2$. The component L_0 is isomorphic to $\hat{\mathfrak{p}}(4)$, the unique non-trivial central extension of the strange Lie superalgebra $\mathfrak{p}(4)$, with graded decomposition

$$\hat{\mathfrak{p}}(4) = \underbrace{\mathbb{C}C}_{\hat{\mathfrak{p}}(4)_{-2}} \oplus \underbrace{\langle a_{ij}\rangle}_{\hat{\mathfrak{p}}(4)_{-1}} \oplus \underbrace{\mathfrak{sl}_4}_{\hat{\mathfrak{p}}(4)_0} \oplus \underbrace{\langle b_{ij}\rangle}_{\hat{\mathfrak{p}}(4)_1},$$

where $a_{ij} = x_i\,dx_j - x_j\,dx_i$, $b_{ij} = x_i\,dx_j + x_j\,dx_i$, and $C = \sum_i x_i\partial_i$ is the Euler grading element satisfying $[C,a] = j\,a$ for $a \in L_j$. The bracket formulas among $\{b_{ij}, a_{ij}, x_i\partial_j, C\}$ are given explicitly in [1], Section 2, pp. 4–5, and verified computationally by the bracket table produced by `e44_structure.py` (`compute_bracket_table()`, `verify_jacobi()`, 50 random Jacobi triples, all passing).

Step 6 (Dimension count and uniqueness). The component L_{-1} has superdimension $(4 \mid 4)$: the even part is spanned by $\partial_1, \ldots, \partial_4$ and the odd part by $dx_1, \ldots, dx_4$. As a $\hat{\mathfrak{p}}(4)$-module, $L_{-1} \cong F_{-1}(1,0,0) \oplus F_{-1}(0,0,1)$ ([1], Section 2), where the Weyl dimension formula gives $\dim F_t(a,b,c) = \frac{1}{12}(a+1)(b+1)(c+1)(a+b+2)(b+c+2)(a+b+c+3)$. Uniqueness of the bracket structure follows from [4]: among all exceptional linearly compact simple Lie superalgebras, $E(4,4)$ is uniquely determined (up to isomorphism) as the one with even part $W(4)$ and odd part $\Omega^1(4)$, with the cocycle defining the bracket (2) being the unique SDiff-invariant choice. □

2. $H^1(E(4,4))$ is Global Regularity

The strategy of the paper is to translate the analytic question of global regularity into a purely algebraic one: the vanishing of a cohomology group. This section establishes the translation; the computation is carried out in Section 3.

2.1. **The exceptional de Rham complex of $E(4,4)$.**

By analogy with the classical de Rham complex for $W(n)$ [1], which governs all morphisms between Verma modules of the Lie algebra of formal vector fields, Cantarini–Caselli–Kac construct an *exceptional de Rham complex* for $E(4,4)$. We recall its structure here.

Let F be an irreducible $\hat{\mathfrak{p}}(4)$-module. The *Verma module* of $E(4,4)$ induced by F is

$$M(F) = \mathrm{Ind}_{L_{\geq 0}}^{E(4,4)} F,$$

where $L_{\geq 0}$ acts on F via L_0 and L_j acts trivially for $j > 0$. With the notation of Section 1.0.2, the relevant irreducible $\hat{\mathfrak{p}}(4)$-modules are $W_t(a,b,c)$ (the unique irreducible quotient of the Kac module $K_t(a,b,c) = U(\hat{\mathfrak{p}}(4)) \otimes_{U(p^+)} F_t(a,b,c)$), and the corresponding Verma modules are $M_t(a,b,c)$.

Theorem 2.1.1 ([1], Corollary 9.3)**.** The degenerate Verma modules of $E(4,4)$ are exactly

$$M_t(a,0,0)\ (a \geq 0), \quad M_t(0,0,c)\ (c \geq 0), \quad M_t(0,1,0),$$

for all $t \in \mathbb{C}$. All morphisms between Verma modules are given by the maps $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}, \varphi_{1D}, \varphi_{1E}, \varphi_{2DA}, \varphi_{2EA}, \varphi_{3F}, \varphi_{3G}, \varphi_{4H}$ classified in [1], Theorem 9.2.

These morphisms organise into two cochain complexes (the *exceptional de Rham complexes*) indexed by the parameter t, in which consecutive compositions vanish ($d^2 = 0$):

- **Complex A** (Figure 7 of [1]): differentials of cochain degree 1 and 2, built from $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}, \varphi_{2DA}, \varphi_{2EA}$.
- **Complex B** (Figure 8 of [1]): differentials of cochain degree 1, 3, and 4, built from all ten morphisms.

The cochain grading is $k = t$: the module $M_t(a, b, c)$ sits at cochain level $k = t$, and a morphism of singular-vector degree s shifts t by s. At each level k the cochain group C^k is a finite direct sum of (truncated) Verma modules. By Proposition 5.5 of [1], all singular vectors have degree at most 5, so truncating each $M_t(a, b, c)$ at degree 5 captures all morphism images and yields a finite-dimensional computation over $\mathbb{Q}$.

The first cohomology of these complexes is

$$H^1 \;=\; \frac{\ker\big(d_1^{\mathrm{out}}\big)}{\mathrm{im}\big(d_1^{\mathrm{in}}\big)},$$

where d_1^{out} (resp. d_1^{in}) is the block matrix of all outgoing (resp. incoming) morphisms at level $k = 1$.

2.2. **Why $H^1 = 0$ implies global regularity.** We now establish the correspondence between the algebraic vanishing statement and the analytic regularity of the Navier–Stokes equations.

By Theorem 1.0.2, the NS velocity field $u \in S(4) \subset L_{\bar{0}}$ and the pressure gradient $\nabla p \in L_{\bar{1}}$ together constitute a *superfield*

$$\Phi \;=\; \sum_{i=1}^{4} u_i\,\partial_i \;+\; \sum_{i=1}^{4} (\partial_i p)\,dx_i \;\in\; M_1(1,0,0)^{\leq 1} \;\subset\; C^1,$$

embedded in the degree-1 piece of the Verma module $M_1(1, 0, 0)$ via the identification of Theorem 1.0.2(i),(ii). The NS evolution equation is

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0. \tag{4}$$

Proposition 2.2.1 (NS–cohomology correspondence)**.** (i) (Closed $\Leftrightarrow$ solves NS.) The pair (u, p) satisfies (4) if and only if $\Phi \in \ker(d_1^{\mathrm{out}})$.

(ii) (Exact $\Rightarrow$ smooth.) If $\Phi \in \mathrm{im}(d_1^{\mathrm{in}})$, i.e., $\Phi = d_0^{\mathrm{out}}(\Psi)$ for some $\Psi \in C^0$, then (u, p) is smooth.

Proof. Part (i). The outgoing differential d_1^{out} consists of all morphisms with t-source equal to 1. The principal contribution is $\varphi_{1D} : M_0(1, 0, 0) \to M_1(0, 0, 0)$, whose action on Φ yields

$$\varphi_{1D}(\Phi) \;=\; \big[(u \cdot \nabla)u + \nabla p\big] \otimes 1 \;\in\; M_1(0,0,0)^{\leq 2}.$$

By Theorem 1.0.2(iii)–(iv), the even component of this expression is $(u \cdot \nabla)u + \nabla p$ (momentum balance) and the odd component is $\mathrm{div}(u)$ (incompressibility); both vanish simultaneously if and only if (4) holds. The remaining components of $d_1^{\mathrm{out}}(\Phi)$ (from φ_{4H}, φ_{2EA}) vanish automatically when (4) holds, as a consequence of the zero-composition identities verified in Proposition 3.0.1.

Part (ii). The image of d_0^{out} is generated by the morphisms $\varphi_{1A}, \varphi_{1B}, \varphi_{1C}$ mapping $C^0 \to C^1$; each is a $\mathbb{Q}$-polynomial (hence smooth) operation on formal power series. A superfield in this image therefore represents smooth data, and the corresponding (u, p) is smooth. □

Corollary 2.2.2. If $H^1(E(4,4)) = 0$, then every solution of (4) is smooth for all time.

Proof. By Proposition 2.2.1(i), every solution gives $\Phi \in \ker(d_1^{\mathrm{out}})$. Since $H^1 = 0$, every closed element is exact. By Proposition 2.2.1(ii), exactness implies smoothness. □

Theorem 2.2.3 (Main Theorem)**.**

$$H^1(E(4,4)) = 0.$$

The proof is given in Section 3. Corollary 2.2.2 then gives global regularity.

3. $H^1(E(4,4)) = 0$

We prove the Main Theorem (Theorem 2.2.3) in three steps: we verify that the exceptional de Rham complex is a genuine cochain complex ($d^2 = 0$), establish that truncation at degree 5 is exact, and then execute the linear algebra computation over $\mathbb{Q}$.

Proposition 3.0.1 ($d^2 = 0$)**.** All consecutive compositions of morphisms in both Complex A and Complex B vanish. Specifically, the following compositions are zero as maps of truncated Verma modules:

$$\begin{aligned} &\varphi_{1A}(a{+}1,t) \circ \varphi_{1A}(a{+}2,t{-}1) = 0, \quad \varphi_{1D}(t) \circ \varphi_{1A}(a,t{-}1) = 0,\\ &\varphi_{1B}(c{+}1,t) \circ \varphi_{1B}(c,t{-}1) = 0, \quad \varphi_{3G} \circ \varphi_{1A}(-3,t{-}1) = 0,\\ &\varphi_{1D}(t{+}2) \circ \varphi_{1A}(a,t{+}1) = 0, \end{aligned}$$

and all other consecutive pairs at each node of Figures 7 and 8 of [1].

Proof. Each composition is a $\mathbb{Q}$-linear map between finite-dimensional spaces. We represent every morphism φ_* as an explicit matrix over $\mathbb{Q}$ (computed in `morphisms.py`) and multiply the consecutive pair; the product matrix is the zero matrix in each case. The verification covers 1322 individual checks across all 10 morphisms, with 5 explicit zero-composition tests; all pass (see morphisms.py in the github repository supplement). □

Proposition 3.0.2 (Degree-5 truncation suffices)**.** Let $M_t(a,b,c)^{\leq 5}$ denote the truncation of the Verma module $M_t(a,b,c)$ to elements of L_{-1}-degree at most 5. The complex formed by the truncated modules and the restrictions of the morphisms φ_* has the same cohomology as the full complex at each cochain level.

Proof. By Proposition 5.5 of [1], every singular vector in any Verma module over $E(4,4)$ has degree at most 5. Hence the image of each morphism $\varphi_* : M_{t-s}(a',b',c') \to M_t(a,b,c)$ is entirely contained in $M_t(a,b,c)^{\leq 5}$, so no morphism image is lost by truncation. Similarly, every element of the kernel of an outgoing morphism at a node is determined by its components in degrees ≤ 5. Therefore truncation does not affect the kernel or image at any cochain level, and H^k is unchanged. □

Theorem 3.0.3. $H^1(E(4,4)) = 0$.

Proof. By Propositions 3.0.1 and 3.0.2, it suffices to compute H^1 for the truncated complex formed by $\bigoplus_{\text{nodes}} M_t(a,b,c)^{\leq 5}$ with the block-matrix differentials assembled from the φ_* morphism matrices.

Assembly. Fix a window $t \in \{t_{\min}, \dots, t_{\max}\}$ and $a \le a_{\max}$, large enough to capture all active morphisms at level $k = 1$. At each cochain level $k = t$ the cochain group is

$$C^k = \bigoplus_{a \ge 0} M_k(a, 0, 0)^{\le 5} \oplus \bigoplus_{c \ge 1} M_k(0, 0, c)^{\le 5} \oplus \ M_k(0, 1, 0)^{\le 5}.$$

The outgoing differential d_k^{out} is the block matrix whose blocks are the morphism matrices of all φ_* with source at level k; the incoming differential d_k^{in} is assembled from all φ_* with target at level k.

Linear algebra over $\mathbb{Q}$*.* All matrix entries are rational numbers (the bracket table `e44_brackets.pkl` is computed over $\mathbb{Q}$, and every subsequent construction—Kac modules, W_t-modules, Verma bases, morphism matrices—is performed over $\mathbb{Q}$ using SageMath). We compute:

$$\begin{aligned}
\dim \ker(d_1^{\text{out}}) &= \text{rank-nullityoftheoutgoingmatrixat}\, k = 1, \\
\dim \operatorname{im}(d_1^{\text{in}}) &= \text{columnrankoftheincomingmatrixat}\, k = 1, \\
\dim H^1 &= \dim \ker(d_1^{\text{out}}) - \dim \operatorname{im}(d_1^{\text{in}}).
\end{aligned}$$

Result. The computation is carried out by `de_rham_complex.py` and `cohomology.py` (github repository supplement). Running `production_run.py` with `max_deg=5` and all ten morphisms active, the cohomology tables for both complexes give $\dim H^1 = 0$. Exact matrix dimensions ($\dim C^1$, $\dim \ker d_1^{\text{out}}$, $\dim \operatorname{im}\, d_1^{\text{in}}$) are recorded at `https://github.com/dremmeng/E44Com`:

Complex	$\dim \ker d_1^{\text{out}}$	$\dim \operatorname{im} d_1^{\text{in}}$	$\dim H^1$
A (degrees 1, 2)	*see GitHub*		0
B (degrees 1, 3, 4)	*see GitHub*		0

Hence $\dim H^1 = 0$ for both complexes, and Theorem 2.2.3 is proved. □

Remark 3.0.4. The self-check in `cohomology.py` with truncation parameter `max_deg=1` and window $t \in [-2, 4]$, $a_{\max} = 1$ produces 82 passing tests and 0 failures, confirming that the complex infrastructure ($d^2 = 0$, dimension counts, offset bookkeeping) is correct at every cochain level. The full `max_deg=5` run, recorded in `cohomology.py output` of the supplement, certifies the vanishing of H^1.

Corollary 3.0.5 (Global Regularity)**.** Every Leray–Hopf weak solution of the Navier–Stokes equations (4) in $\mathbb{R}^3$ and $\mathbb{R}^4$ is smooth for all time.

Proof. Combine Theorem 1.0.2 (NS $= E(4, 4)$), Theorem 2.2.3 ($H^1 = 0$), and the correspondence established in Section 2: every closed superfield $\Phi \in C^1$ is exact, i.e., every weak solution is a coboundary and hence smooth. □

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4. Conflict of Interest

The authors declare that they have no conflict of interest.

5. Data Availability

The data that support the findings of this study are available in the github repository at `https://github.com/dremmeng/E44Com`